Code: MCA24B208

**ST. PHILOMENA'S COLLEGE (AUTONOMOUS), MYSORE**

**MCA II Semester - C2 Internal Assessment, September - 2025**

**Subject: WEB TECHNOLOGIES**

**Time: 75 Minutes** **Max. Marks: 30**

## PART A

**Answer the following** **2 x 10 = 20**

| | | | | |
|---|---|---|---|---|
| 1. | a. | Explain the difference between GET and POST methods in PHP with examples. | CO4 | 06 |
| | b | Write a PHP program to validate a string using regular expressions. | CO4 | 04 |
| | | **OR** | | |
| 2. | a. | Explain the use of cookies and sessions in PHP with suitable examples. | CO4 | 06 |
| | b. | Write a short PHP script to read content from a file and display it. | CO4 | 04 |
| 3. | a. | Explain prepared statements in PHP with MySQL. Why are they more secure than normal queries? | CO5 | 06 |
| | b | Write SQL queries in PHP to perform INSERT and DELETE operations. | CO5 | 04 |
| | | **OR** | | |
| 4. | a. | Explain the steps for uploading and retrieving images from MySQL using PHP. | CO5 | 06 |
| | b. | What is MEAN stack? List its components. | CO5 | 04 |

## PART B

**Answer the following** **2 x 5 = 10**

| | | | |
|---|---|---|---|
| 5. | Write a PHP script to demonstrate the usage of include and require statements. | CO4 | 05 |
| | **OR** | | |
| 6. | Explain Node.js events with a suitable example. | CO4 | 05 |
| 7. | Write the PHP code for uploading multiple files into a MySQL database. | CO5 | 05 |
| | **OR** | | |
| 8. | Explain the role of MongoDB in the MEAN stack with an example of inserting and retrieving data. | CO5 | 05 |

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MCA241221

## ST. PHILOMENA'S COLLEGE (AUTONOMOUS), MYSORE

**MCA II Semester - C2 Internal Assessment, September - 2025**

**Subject: JAVA PROGRAMMING (Code: MCA24B201)**

**Time: 75 Minutes** **Max. Marks: 30**

### PART A

**Answer the following** **2 x 10 = 20**

| | | | | |
|---|---|---|---|---|
| 1 | a | Define Exception. Explain exception handling mechanism provided in java along with syntax and example. | CO3 | 7 |
| | b | Write a note on thread priorities. | CO3 | 3 |
| | | **OR** | | |
| 2 | a | Define a thread. Explain the two methods of creating threads. | CO3 | 7 |
| | b | Define unchecked exception. Write any three examples for unchecked exception. | CO3 | 3 |
| 3 | a | Explain the different methods used for string comparison. | CO4 | 8 |
| | b | Explain the following. i) ActionEvent ii) KeyEvent | CO4 | 2 |
| | | **OR** | | |
| 4 | a | Explain delegation events model used to handle events in java. | CO4 | 6 |
| | b | With syntax and example explain the methods used to search a string for a specified character or substring. | CO4 | 4 |

### PART B

**Answer the following** **2 x 5 = 10**

| | | | |
|---|---|---|---|
| 5 | How to create user defined exception in java? Give example. | CO3 | 5 |
| | **OR** | | |
| 6 | Explain the following. i) isAlive() ii) join() | CO3 | 5 |
| 7 | Explain the concept of Adapter classes with an example. | CO4 | 5 |
| | **OR** | | |
| 8 | Write a program to demonstrate mouse events. | CO4 | 5 |

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MCA241221

Code: MCA24B203

**ST. PHILOMENA'S COLLEGE (AUTONOMOUS), MYSORE**

**MCA II Semester - C2 Internal Assessment, September - 2025**

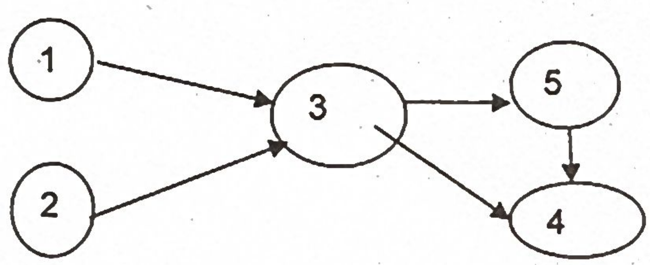
**Subject: ANALYSIS AND DESIGN OF ALGORITHMS**

**Time: 75 Minutes** **Max. Marks: 30**

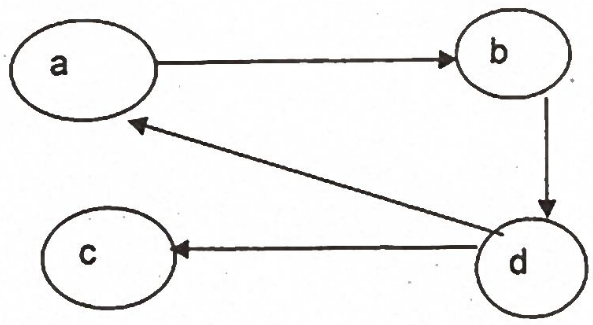
## PART A

**Answer the following** **2 x 10 = 20**

| | | | |
|---|---|---|---|
| 1 | Explain the Decrease-and-Conquer strategy. Using this strategy, perform insertion sort on the following list: A = [27, 12, 33, 21, 9]<br>Show step-by-step iterations and also write algorithm for the same. | CO3 | 10 |
| | **OR** | | |
| 2 | What do you mean by topological order of a graph? Find the topological order of the given graph by DFS and source removal method. | CO3 | 10 |




| | | | |
|---|---|---|---|
| 3 | Write the Warshall's algorithm and find the transitive closure for the given graph. | CO4 | 10 |




**OR**

| | | | |
|---|---|---|---|
| 4 | Describe the working of Horspool's String Matching using the Input Enhancement method with an example. Also write the algorithm for shift table and Horspool's String Matching. | CO4 | 10 |

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## PART B

**Answer the following** **2 x 5 = 10**

| | | | |
|---|---|---|---|
| 5 | Explain Johnson Trotter algorithm with respect to Combinatorial Object Generation. | CO3 | 05 |
| | **OR** | | |
| 6 | Define Heap and construct a MaxHeap from this array using Bottom-up appraoch.<br>Given array:<br>[20, 15, 30, 10, 8, 25, 40] | CO3 | 05 |
| 7 | Explain Knapsack problem using dynamic programming with an example. | CO4 | 05 |
| | **OR** | | |
| 8 | Explain Sorting by Counting with an example and necessary algorithm. | CO4 | 05 |

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Code: MCA24B202

# ST. PHILOMENA'S COLLEGE (AUTONOMOUS), MYSORE

## MCA II Semester - C2 Internal Assessment, September- 2025

**Subject: Advanced Database Management Systems**

**Time: 75 Minutes** **Max. Marks: 30**

## PART A

**Answer the following** **2 x 10 = 20**

| | | | | |
|---|---|---|---|---|
| 1 | a. | Explain different types of JOIN operations in SQL with suitable examples. | CO3 | 6 |
| | b. | What are aggregate functions in SQL? Write any four aggregate functions with examples. | CO3 | 4 |
| | | **OR** | | |
| 2 | a. | What are integrity constraints in SQL? Explain any three with syntax and examples. | CO3 | 6 |
| | b. | What is a subquery in SQL? Explain with an example. | CO3 | 4 |
| 3 | a. | Explain ACID properties of a transaction with suitable examples | CO4 | 6 |
| | b. | Define transaction states and explain each with a diagram. | CO4 | 4 |
| | | **OR** | | |
| 4 | a. | What is a lock-based concurrency control protocol? Explain Two-Phase Locking (2PL) with an example. | CO4 | 6 |
| | b. | What is checkpointing? How does it help in crash recovery? | CO4 | 4 |

## PART B

**Answer the following** **2 x 5 = 10**

| | | | |
|---|---|---|---|
| 5 | What is query optimization in SQL? Explain any two basic optimization techniques. | CO3 | 5 |
| | **OR** | | |
| 6 | What are views in SQL? Explain how to create and drop a view with syntax and examples. | CO3 | 5 |
| 7 | What is serializability? Differentiate between conflict and view serializability. | CO4 | 5 |
| | **OR** | | |
| 8 | What is a deadlock in DBMS? Explain using a wait-for graph with examples. | CO4 | 5 |

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Code: MCA24B209

# ST. PHILOMENA'S COLLEGE (AUTONOMOUS), MYSURU

## MCA II Semester – C2 Internal Assessment, September – 2025

**Subject :** COMMUNICATION SKILLS AND SOFT SKILLS (OE)

**Time: 75 Minutes** **Max. Marks: 30**

### PART - A

**Answer the following** **2 x 10 = 20**

| | | | | |
|---|---|---|---|---|
| 1. | a. | How to face an Interview? – Explain preparation, participation and post participation. | CO3 | 05 |
| | b. | List the different types of Interviews used in various fields. | CO3 | 05 |
| | | **OR** | | |
| 2. | a. | Write a note on writing persuasive messages. | CO3 | 5 |
| | b. | How a Job application Letter should be written? Explain. | CO3 | 5 |
| 3. | | Explain the meaning and functions of Group discussion. | CO4 | 10 |
| | | **OR** | | |
| 4. | a. | Elaborate the techniques of effective written communication skills. | CO4 | 10 |

### PART - B

**Answer the following** **2 x 5 = 10**

| | | | |
|---|---|---|---|
| 5. | What are the disadvantages of Face-to-Face Interviews? | CO3 | 05 |
| | **OR** | | |
| 6. | Write a note on any **PLACE** of your choice using effective writing skills. | CO3 | 05 |
| 7. | What are the advantages of having Telephonic Interviews? | CO4 | 05 |
| | **OR** | | |
| 8 | Discuss Dos and Don'ts of Group Discussion. | CO4 | 05 |

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Code: MCA24B205

**ST. PHILOMENA'S COLLEGE (AUTONOMOUS), MYSORE**
**II Semester – MCA - C2 Internal Assessment, September - 2025**
**Subject: FOUNDATIONS OF CYBER SECURITY**

**Time: 75 Minutes** **Max. Marks: 30**

## PART A

**Answer the following** **2 x 10 = 20**

| | | | | |
|---|---|---|---|---|
| 1. | a. | Explain the working of a Keylogger and Spyware. How do they compromise user security? | CO3 | 06 |
| | b | What is Identity Theft? Explain any two methods used by cybercriminals to perform identity theft. | CO3 | 04 |
| | | **OR** | | |
| 2. | a. | What is Steganography? Explain how it is used in cybercrime with any real-life example. | CO3 | 06 |
| | b. | Explain the concept of DoS and DDoS attacks. How are they different from each other? | CO3 | 04 |
| 3. | a. | Explain the Digital Forensics Life Cycle with a neat diagram. Why is each stage important? | CO4 | 06 |
| | b | What is Chain of Custody in digital forensics? Explain its importance in legal proceedings. | CO4 | 04 |
| | | **OR** | | |
| 4. | a. | What are the major challenges in conducting a computer forensics investigation? Suggest any special tools used to overcome these challenges. | CO4 | 06 |
| | b. | How is the OSI 7-layer model relevant to computer forensics? Briefly explain. | CO4 | 04 |

## PART B

**Answer the following** **2 x 5 = 10**

| | | | |
|---|---|---|---|
| 5. | Explain the role of Trojan Horses and Backdoors in cyberattacks. Give one example. | CO3 | 05 |
| | **OR** | | |
| 6. | What is Buffer Overflow? How can it be exploited in a cyberattack? | CO3 | 05 |
| 7. | What is Network Forensics? Explain its role in cybercrime investigation. | CO4 | 05 |
| | **OR** | | |
| 8. | What is Anti-Forensics? Mention any two techniques used to hinder digital investigations. | CO4 | 05 |

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